

1. Logic Gates

Design Logic

The logic gates were implemented using Verilog continuous assignment statements.

- AND Gate: Output is HIGH only when both inputs are HIGH.
- OR Gate: Output is HIGH when at least one input is HIGH.
- NOT Gate: Produces the complement of the input.
- NAND Gate: Inverse of AND operation.
- NOR Gate: Inverse of OR operation.
- XOR Gate: Output is HIGH when inputs are different.
- XNOR Gate: Output is HIGH when inputs are the same.

Verilog statements used:

```
assign AND_OUT = A & B;
```

```
assign OR_OUT = A | B;
```

```
assign NOT_OUT = ~A;
```

```
assign NAND_OUT = ~(A & B);
```

```
assign NOR_OUT = ~(A | B);
```

```
assign XOR_OUT = A ^ B;
```

```
assign XNOR_OUT = ~(A ^ B);
```

Simulation Results

All possible combinations of inputs A and B were applied.

A	B	AND	OR	NOT(A)	NAND	NOR	XOR	XNOR
---	---	-----	----	--------	------	-----	-----	------

0	0	0	0	1	1	1	0	1
---	---	---	---	---	---	---	---	---

0	1	0	1	1	1	0	1	0
---	---	---	---	---	---	---	---	---

1	0	0	1	0	1	0	1	0
---	---	---	---	---	---	---	---	---

1	1	1	1	0	0	0	0	1
---	---	---	---	---	---	---	---	---

The simulation outputs matched the expected truth tables, verifying the correct operation of all logic gates.

```
HP@LAPTOP-1I469TMH MINGW64 ~
$ cd /d/verilogcodes

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ iverilog -o logicGates.vvp logicGates.v

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ iverilog -o logicGatesTB.vcd logicGatesTB.v logicGates.v

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ vvp logicGates.vcd
logicGates.vcd: Unable to open input file.

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ vvp logicGatesTB.vcd
VCD info: dumpfile logic_gates.vcd opened for output.
A B | AND OR NOT NAND NOR XOR XNOR
0 0 | 0 0 1 1 1 0 1
0 1 | 0 1 1 1 0 1 0
1 0 | 0 1 0 1 0 1 0
1 1 | 1 1 0 0 0 0 1
logicGatesTB.v:60: $finish called at 40000 (1ps)

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ |
```

2. Half Adder

Design Logic

A Half Adder performs the addition of two single-bit binary inputs A and B.

It produces:

- SUM: Result of addition
- CARRY: Carry generated during addition

Logic equations:

[SUM = A xor B]

[CARRY = A . B]

The SUM output is generated using an XOR gate, while the CARRY output is generated using an AND gate.

Verilog implementation:

assign SUM = A ^ B;

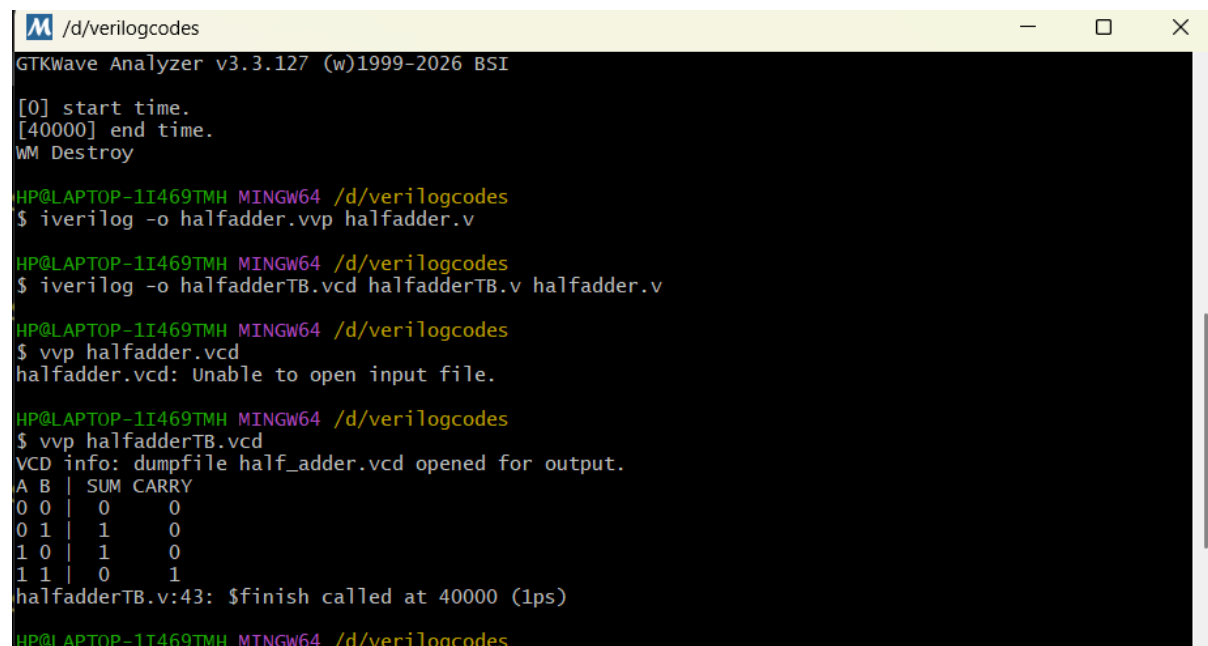
assign CARRY = A & B;

Simulation Results

A B SUM CARRY

0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

The simulation confirms that the Half Adder correctly adds two binary digits and generates the appropriate carry output.



```
/d/verilogcodes
GTKWave Analyzer v3.3.127 (w)1999-2026 BSI

[0] start time.
[40000] end time.
WM Destroy

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ iverilog -o halfadder.vvp halfadder.v

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ iverilog -o halfadderTB.vcd halfadderTB.v halfadder.v

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ vvp halfadder.vcd
halfadder.vcd: Unable to open input file.

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ vvp halfadderTB.vcd
VCD info: dumpfile half_adder.vcd opened for output.
A B | SUM CARRY
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1
halfadderTB.v:43: $finish called at 40000 (1ps)

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
```

3. Full Adder

Design Logic

A Full Adder performs the addition of three one-bit inputs:

- A
- B
- Carry Input (CIN)

It generates:

- SUM
- Carry Output (COUT)

Logic equations:

[SUM = A xor B xor CIN]

[COUT = AB + BCIN + ACIN]

The SUM output becomes HIGH when an odd number of inputs are HIGH. The COUT output becomes HIGH whenever two or more inputs are HIGH.

Verilog implementation:

```
assign SUM = A ^ B ^ CIN;
```

```
assign COUT = (A & B) | (B & CIN) | (A & CIN);
```

Simulation Results

A	B	CIN	SUM	COUT
---	---	-----	-----	------

0	0	0	0	0
---	---	---	---	---

0	0	1	1	0
---	---	---	---	---

0	1	0	1	0
---	---	---	---	---

0	1	1	0	1
---	---	---	---	---

1	0	0	1	0
---	---	---	---	---

1	0	1	0	1
---	---	---	---	---

1	1	0	0	1
---	---	---	---	---

1	1	1	1	1
---	---	---	---	---

The simulation results match the theoretical truth table of a Full Adder, confirming correct implementation of binary addition with carry propagation.

```
HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ iverilog -o fulladder.vvp fulladder.v

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ iverilog -o fulladderTB.vcd fulladderTB.v fulladder.v

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ vvp fulladder.vcd
fulladder.vcd: Unable to open input file.

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$ vvp fulladderTB.vcd
VCD info: dumpfile full_adder.vcd opened for output.
A B CIN | SUM COUT
0 0 0 | 0 0
0 0 1 | 1 0
0 1 0 | 1 0
0 1 1 | 0 1
1 0 0 | 1 0
1 0 1 | 0 1
1 1 0 | 0 1
1 1 1 | 1 1
fulladderTB.v:41: $finish called at 80000 (1ps)

HP@LAPTOP-1I469TMH MINGW64 /d/verilogcodes
$
```

Conclusion

The Verilog designs for basic logic gates, Half Adder, and Full Adder were successfully implemented and simulated using Icarus Verilog and GTKWave. The observed outputs matched the expected truth tables for all input combinations, validating the correctness of the designs.

